Models of Curves and Valuations

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Preface

This is a preliminary version of an open book developed as part of the MCLF project.

The text is work in progress. Individual versions should be cited by referring to a specific Git commit or release.

Chapter 1

Valued Fields and Models

1.1 Models of Curves

1.1.1 Basic Definitions

Let K be a field endowed with an additive valuation of rank 1, $v_K : K \to \hat{\mathbb{R}} := \mathbb{R} \cup \{\infty\}$. Let $\Gamma_K := v_K(K^*) \subset \mathbb{R}$ be the value group of v_K, $\mathcal{O}_K \subset K$ its valuation ring with maximal ideal $\mathfrak{m}_K$ and $k := \mathcal{O}_K/\mathfrak{m}_K$ its residue field. We assume that v_K is discrete, K is complete with respect to v_K and k is perfect. Let X be a smooth, projective and absolutely irreducible curve over K.

Definition 1.1. A *model* $\mathcal{X}$ of X is a proper, flat and normal $\mathcal{O}_K$-scheme with an identification of its generic fibre $\mathcal{X}_g$ with X.

Example 1.2. Let $K = \mathbb{Q}_p$, $\mathcal{O}_K = \mathbb{Z}_p$ and $X = \mathbb{P}^1(K)$, then $\mathcal{X} = \mathbb{P}^1(\mathcal{O}_K)$ is a model of X with the evident isomorphism from $\mathcal{X}_g = \mathcal{X} \times_{\mathcal{O}_K} K$ to $\mathbb{P}^1(K)$. The special fibre[1] $\mathcal{X}_s$ has one irreducible component which is isomorphic to $\mathbb{P}^1(k)$.

Remark 1.3. The condition of a model being flat is equivalent to it being integral: If $\mathcal{X} \to \operatorname{Spec} \mathcal{O}_K$ is an integral scheme, then for every $x \in \mathcal{X}$, the stalk $\mathcal{O}_{\mathcal{X},x}$ is an integral ring extension of $\mathcal{O}_K$. Thus $\mathcal{O}_{\mathcal{X},x}$ is a torsion-free module over the discrete valuation ring $\mathcal{O}_K$, hence flat by [**?**, Corollary 1.2.14]. Conversely, the integrality of a model $\mathcal{X}$ of X follows from its flatness and the integrality of X by [**?**, Proposition 4.3.8].

Remark 1.4. A point $x \in \mathcal{X}$ of codimension 1 which is regular is normal. This is conversely true by Serre's criterion (see [**?**, Lemma 8.2.21]).

Example 1.5. Consider the curve X defined by $y^2 = x^3 + px$ over $\mathbb{Q}_p$ with $p \neq 2$. We show that

$$\mathcal{X} := \operatorname{Proj} \mathbb{Z}_p[X, Y, Z]/(Y^2 Z - X^3 - pXZ^2) \tag{1.6}$$

[1] That is, the fibre of the closed point of $\operatorname{Spec} \mathcal{O}_K$.

is a model of X. It is clearly a proper scheme over $\mathcal{O}_K = \mathbb{Z}_p$. For the flatness it suffices to show that $R := \mathbb{Z}_p[X,Y,Z]/(Y^2Z - X^3 - pXZ^2)$ is a torsion-free $\mathbb{Z}_p$-module by Remark 1.3. But this is true since $Y^2Z - X^3 - pXZ^2$ is irreducible in $\mathbb{Z}_p[X,Y,Z]$.
For normality, consider the affine patch $U : Z \neq 0$, that is

$$U := \operatorname{Spec} \mathbb{Z}_p[x,y]/(y^2 - x^3 + px) \tag{1.7}$$

where $x = X/Z$ and $y = Y/Z$. The special fibre $\mathcal{X}_s$ is defined by the equation $\bar{X}_p : \bar{y}^2 - \bar{x}^3 = 0$, which is irreducible in $\mathbb{F}[x,y]$ Let Z be the irreducible component of $\mathcal{X}_s$ containing the generic point η. The only singularity is when $\eta : y = x = 0$ and so one has the maximal ideal $\mathfrak{m}_p = (x,y,p)$ of $\mathcal{O}_{Z,\eta}$. All monomials have degrees greater than 2, so $\mathfrak{m}_p$ can't be generated by less than three elements modulo $\mathfrak{m}_p^2$. This implies

$$\dim_{\mathbb{F}_p} \mathfrak{m}_p/\mathfrak{m}_p^2 = 3 > 2 = \operatorname{codim} \eta, \tag{1.8}$$

i.e. η is a non-regular point of U.

1.1.2 From Models to Valuations

In this section, we assume the valuation v_K to be *normalised*, i.e. $v_K(K^*) = \mathbb{Z}$.

Definition 1.9. Let F_X be a function field of X over K. A valuation $v : F_X \to \hat{\mathbb{R}}$ is said to be *of type II*, if

1. $v|_K = v_K$,
2. v is discrete, and
3. the extension of residue fields $k(v)|k$ has transcendence degree 1.

A valuation of this type is also known as *geometric valuation*. We write $V_{\mathrm{II}}(F_X)$ for the set of all valuations of type II of F_X. The number

$$m(v) := [\Gamma_v : \Gamma_K] \tag{1.10}$$

is called the *multiplicity of* v, where $\Gamma_v := v(F_X^*)$ is the value group. We say that v is *weakly unramified*, if $m(v) = 1$.

Example 1.11. The *Gauß valuation* on the polynomial ring $K[x]$ is defined as

$$v_g : K[x] \to \hat{\mathbb{Q}},\ v_g\left(\sum_i a_i x^i\right) = \min_i(v_K(a_i)). \tag{1.12}$$

Since $k(v_g) = k(\bar{x})$ (where $\bar{x}$ is the image of x in $k(v_g)$), the Gauß valuation extends naturally and uniquely to $K(x)$, meaning that we have $v_g \in V_{\mathrm{II}}(K(x))$. We will later see that $\{v_g\}$ defines the model $\mathcal{X} = \mathbb{P}^1(\mathbb{Z}_p)$ of $\mathbb{P}^1(\mathbb{Q}_p)$.

Let $\mathcal{X}$ be a model of X. We wish to assign a valuation in $V_{\mathrm{II}}(F_X)$ to each generic point of the special fibre $\mathcal{X}_s$. Suppose that Z is an irreducible component of $\mathcal{X}_s$ with generic point ξ. Since ξ has codimension 1 in $\mathcal{X}$, the stalk $\mathcal{O}_{\mathcal{X},\xi}$ is a discrete valuation ring. Thus, we can define a valuation on F_X,

$$v_Z : \mathrm{Frac}(\mathcal{O}_{\mathcal{X},\xi}) = F_X \to \hat{\mathbb{Q}}. \tag{1.13}$$

We set

$$V(\mathcal{X}) := \{v_Z : \ Z \subset \mathcal{X}_s \text{ is an irreducible component}\}. \tag{1.14}$$

Proposition 1.15. *The valuation defined in* (1.13) *is of type II.*

Proof. Since $\mathcal{X}$ is a flat $\mathcal{O}_K$-scheme $\mathcal{O}_{\mathcal{X},\xi}$ dominates $\mathcal{O}_K$, and we may scale v_Z such that the restriction of v_Z to K is v_K. Furthermore, we have a closed immersion $Z \hookrightarrow \mathcal{X}$ by definition, inducing a surjection $\mathcal{O}_{\mathcal{X},\xi} \to \mathcal{O}_{Z,\xi}$. Thus $\mathrm{trdeg}(k(v_Z)) = \mathrm{codim}(\mathcal{O}_{Z,\xi}) = \mathrm{codim}(v_Z) = 1$, showing $v_Z \in V_{\mathrm{II}}(F_X)$. □

Definition 1.16. The *multiplicity* of the irreducible component Z of $\mathcal{X}_s$ containing the generic point ξ is defined as

$$m(Z) := \mathrm{length}(\mathcal{O}_{\mathcal{X},\xi}). \tag{1.17}$$

Proposition 1.18. *It holds* $m(v_Z) = m(Z)$.

Proof. As the model $\mathcal{X}$ is normal, the stalk $\mathcal{O}_{\mathcal{X},\xi}$ is a discrete valuation ring. Let π_ξ denote an uniformizing parameter, then

$$\mathcal{O}_{\mathcal{X}_s,\xi} = \mathcal{O}_{\mathcal{X},\xi}/(\pi_\xi). \tag{1.19}$$

Thus, the length of the stalk at ξ coincides with the length of the chain of ideals generated by the uniformizing parameter of the quotient ring: Since $(\pi_\xi) = (\pi_{Z,\xi}^m)$, where $m = (\Gamma_{v_Z} : \Gamma_K)$, it holds $m(v_Z) = \mathrm{length}(\mathcal{O}_{\mathcal{X},\xi})$ as desired. □

Definition 1.20. A *modification* of a model of X is a morphism $\mathcal{X}' \to \mathcal{X}$ of $\mathcal{O}_K$-schemes which induces the identity on the generic fibre.

Remark 1.21. Since a model is proper over $\mathcal{O}_K$, a modification is a proper morphism. Thus since it is dominant, it is also surjective.

The main goal of the first two sections of this chapter is to establish the following theorem.

Theorem 1.1. *The map*

$$\mathcal{X} \mapsto V(\mathcal{X}). \tag{1.22}$$

defines a monotone bijection between

- *the set of isomorphism classes of models of X, and*
- *the set of finite, nonempty subsets of $V_{II}(F_X)$.*

In the remainder of this section, we prove injectivity of this map.

Lemma 1.23 ([?, Lemma 3.6])**.** *Let $\mathcal{X}$ be a model of X and let $\mathcal{X}'$ be a model of another irreducible smooth projective curve X'. Let $\varphi : \mathcal{X}' \to \mathcal{X}$ be a morphism of $\mathcal{O}_K$-schemes which induces a cover $X' \to X$ on the generic fibre. Let η be a generic point of $\mathcal{X}_s$, then the preimage of η under φ is non-empty and consists of generic points of $\mathcal{X}'_s$. If φ is birational, then the preimage consists of precisely one generic point.*

Proof. Let $Z \subset \mathcal{X}_s$ be the irreducible component that contains η. Since φ is surjective (cf. Remark 1.21), there is a point η' on an irreducible component $Z' \subset \mathcal{X}'_s$ which is mapped to η. Since φ is dominant, the induced ring homomorphism $\mathcal{O}_{\mathcal{X},\eta} \to \mathcal{O}_{\mathcal{X}',\eta'}$ is injective. This is an extension of discrete valuation rings and thus $\eta' \in Z'$ must be generic. In particular, if φ is birational, then both rings are integrally closed in the same field of fractions which makes them equal. Therefore, η' must be unique since $\mathcal{X}'$ is separated. $\square$

Lemma 1.24 ([?, Lemma 3.7])**.** *Let $\varphi : \mathcal{X}' \to \mathcal{X}$ be a modification of models of X. Then $V(\mathcal{X}) \subset V(\mathcal{X}')$.*

Proof. This is a consequence of Proposition 1.15 and Lemma 1.23. $\square$

We have a partial order of models of X given by $\mathcal{X} \leq \mathcal{X}'$ if there exists a modification $\mathcal{X}' \to \mathcal{X}$. By the previous lemma, this gives a partial order on the corresponding finite subsets of $V_{\mathrm{II}}(F_X)$ via inclusion maps.

Lemma 1.25 ([?, Lemma 3.8])**.** *Let $\mathcal{X}$ and $\mathcal{X}'$ be models of X with $V(\mathcal{X}) \subset V(\mathcal{X}')$. Then there is a modification $\mathcal{X}' \to \mathcal{X}$.*

Sketch of the proof. TBD.

$\square$

Proposition 1.26 ([?, Proposition 3.9],[?, Corollary 8.3.23])**.** *Two models $\mathcal{X}$ and $\mathcal{X}'$ of X are isomorphic as models of X if and only if $V(\mathcal{X}) = V(\mathcal{X}')$. In particular, the map defined in (1.22) is injective.*

Proof. Suppose that $V(\mathcal{X}) = V(\mathcal{X}')$. Then there exist morphisms $\mathcal{X} \to \mathcal{X}'$ and $\mathcal{X}' \to \mathcal{X}$. The composition of these morphisms belongs to the equivalence class of the identity morphism, when considered as a rational map. $\square$

Example 1.27. We continue with Example 1.5 of the model

$$\mathcal{X} = \operatorname{Proj} \mathbb{Z}_p[X, Y, Z]/(Y^2 Z - X^3 - pXZ^2)$$

of the elliptic curve $X: \ y^2 = x^3 + px$ defined over $\mathbb{Q}_p$. We have already checked that it is a model. Let Z be the irreducible component in $\mathcal{X}_s$ containing the generic point ξ and denote the corresponding valuation by v_Z. Z is defined by $\bar{X}_p: \ y^2 = x^3$, thus $Z = \mathcal{X}_s$ and $V(\mathcal{X}) = \{v_Z\}$.
We want to describe v_Z as

$$v_Z|_{F_X} = v_g$$

with respect to the x-coordinate. Using Lemma 1.43, we see that $v_Z(p) = 1$ and $v_Z(x) = 0$, since the horizontal divisor on $x = 0$ intersects $\bar{X}_p$ at a unique point

1.2 From Valuations to Models

For the proof of Theorem 1.1, it remains to show the surjectivity of the map ((1.22)), i.e., for a given finite, non-empty set of valuations $V \subset V_{\mathrm{II}}(F_X)$, there is a model $\mathcal{X}$ of X such that $V = V(\mathcal{X})$. We will prove this in two steps, which we state in the following proposition.

Proposition 1.28 ([?, 3.14, 3.16], [?, 8.3.36])**.** *(i) Given model $\mathcal{X}$ and a valuation $v \in V_{II}(F_X)$, there exists a modification $\varphi\colon \mathcal{X}' \to \mathcal{X}$ such that $v \in V(\mathcal{X}')$.*

(ii) Given a model $\mathcal{X}$ and a nonempty subset $V \subset V(\mathcal{X})$, there exists a modification $\varphi\colon \mathcal{X} \to \mathcal{X}'$ such that $V = V(\mathcal{X}')$.

After proving Proposition 1.28, the proof is complete: Starting with a model $\mathcal{X}$ of X, we can apply part (i) for every $v \in V \subset V_{\mathrm{II}}(F_X)$ to obtain a new model that induces all valuations in V, but possibly more. We then apply part (ii) to construct a model that induces exactly the valuations in V.

1.2.1 Proof of the First Part of Theorem 1.1

We fix a model $\mathcal{X}$ of X and a type II valuation $v \in V_{\mathrm{II}}(F_X)$.

Definition 1.29. A point $x \in \mathcal{X}$ is called a *centre* of v if there exists a morphism $f\colon \operatorname{Spec}\mathcal{O}_v \to \mathcal{X}$ of $\mathcal{O}_K$-schemes which maps the closed point to x and makes the following diagram commute:

$$\begin{tikzcd}
\operatorname{Spec}\mathcal{O}_v \arrow[rr, "f"] & & \mathcal{X} \\
& \operatorname{Spec} F_X \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

Here, the left map is induced by the inclusion $\mathcal{O}_v \hookrightarrow \operatorname{Frac}\mathcal{O}_v = F_X$, and the right map comes from the identification of X with the special fibre $\mathcal{X}_s$.

Specifically, it is the canonical map $\operatorname{Spec} F_X \cong \operatorname{Spec} \mathcal{O}_{\mathcal{X},\xi} \to \mathcal{X}$, where ξ is the generic point of $\mathcal{X}$.

Lemma 1.30. *There exists a unique centre $x \in \mathcal{X}$ of v.*

Proof. By the valuative criterion for properness, such a centre exists, and the morphism f is unique (see also [**?**, Section 01J5, Lemma 01J6]):

$$\begin{tikzcd} \operatorname{Spec} F_X \arrow[r] \arrow[d] & \mathcal{X} \arrow[d] \\ \operatorname{Spec} \mathcal{O}_v \arrow[r] \arrow[ur, dotted, "f"] & \operatorname{Spec} \mathcal{O}_K . \end{tikzcd}$$

□

Lemma 1.31 ([**?**, 3.11])**.** *We have $v \in V(\mathcal{X})$ if and only if the centre x of v is of codimension one. In that case x is the generic point on the special fibre which induces v.*

Proof. Assume $v \in V(\mathcal{X})$. Then v is induced by a generic point $\eta \in \mathcal{X}_s$ on the special fibre. The canonical morphism $\operatorname{Spec} \mathcal{O}_{\mathcal{X},\eta} \to \mathcal{X}$ shows that v has centre η, which is indeed a point of codimension one. Conversely, suppose $v \in V_{\mathrm{II}}(F_X)$ has center $x \in \mathcal{X}$ of codimension one. The morphism $\operatorname{Spec} \mathcal{O}_v \to \mathcal{X}$ uniquely factors through $\operatorname{Spec} \mathcal{O}_v \to \operatorname{Spec} \mathcal{O}_{\mathcal{X},x}$, so $\mathcal{O}_v$ dominates $\mathcal{O}_{\mathcal{X},x}$. Since these are both valuation rings with common fraction field K, it follows that they are in fact equal. Thus $v \in V(\mathcal{X})$. □

Lemma 1.32 ([**?**, 3.12, 3.13])**.** *Let $x \in \mathcal{X}$ be the centre of v, and let $f \in \mathcal{O}_v^* \subset F_X$ be such that the reduction $\overline{f} \in k(v)$ is transcendental over k. Then:*

(a) If $f \in \mathcal{O}_{\mathcal{X},x}$, then $v \in V(\mathcal{X})$

(b) If the rational map $\mathcal{X} \dashrightarrow \mathbb{P}^1(\mathcal{O}_K)$ induced by f is defined at the center x of v, then $v \in V(\mathcal{X})$.

Proof. (a) By Lemma 1.31, it suffices to show that x is a point of codimension one. Choose an open affine set $\operatorname{Spec} A = U \subset \mathcal{X}$ such that $x \in U$ and $f \in A$. Denote by $\mathfrak{p} \subset A$ the prime ideal corresponding to x. Since v is non-trivial, the center x is not the generic point of $\mathcal{X}$, and so the codimension is not zero. If the dimension of $A/\mathfrak{p}$ is at least one, then the inequality

$$\dim_(A_{\mathfrak{p}}) + \dim_(A/\mathfrak{p}) \leq \dim_(A) = 2$$

implies

$$\operatorname{codim}(\overline{\{x\}}, \mathcal{X}) = \dim_(A_{\mathfrak{p}}) \leq 2 - \dim_(A/\mathfrak{p}) \leq 2 - 1 = 1,$$

and hence equals one. Thus, it suffices to show that $A/\mathfrak{p}$ is not a field. If $A/\mathfrak{p}$ were a field, it would be a finite extension of k, but this is impossible since $\overline{f} \in A/\mathfrak{p} \subset k(v)$ is transcendental over k.

(b) It suffices to show, using (a), that $f \in \mathcal{O}_{\mathcal{X},x}$. Let $P = V(f^{-1})$ and $Z = V(f)$ be the sets of poles and zeros of f, respectively. Then the rational map $\mathcal{X} \dashrightarrow \mathbb{P}^1(\mathcal{O}_K)$ is obtained by gluing the morphisms

$$\mathcal{X}\backslash P \to \operatorname{Spec} \mathcal{O}_K[t] \subset \mathbb{P}^1(\mathcal{O}_K), \quad \text{and} \quad \mathcal{X}\backslash Z \to \operatorname{Spec} \mathcal{O}_K[t^{-1}] \subset \mathbb{P}^1(\mathcal{O}_K),$$

which are induced by

$$\begin{aligned} \mathcal{O}_K[t] &\to \Gamma(\mathcal{X} \setminus P, \mathcal{O}_{\mathcal{X}}), \\ t &\mapsto f \end{aligned} \quad \text{and} \quad \begin{aligned} \mathcal{O}_K[t^{-1}] &\to \Gamma(\mathcal{X} \setminus Z, \mathcal{O}_{\mathcal{X}}), \\ t^{-1} &\mapsto f^{-1} \end{aligned}$$

and is defined on $U := (\mathcal{X} \setminus P) \cup (\mathcal{X} \setminus Z) = \mathcal{X} \setminus (P \cap Z)$. By assumption, $x \in U$. If $x \in \mathcal{X} \setminus P$, the claim follows immediately, as

$$f \in \Gamma(\mathcal{X} \setminus P, \mathcal{O}_{\mathcal{X}}) \subset \mathcal{O}_{\mathcal{X},x}.$$

If $x \in \mathcal{X} \setminus Z$, then

$$f^{-1} \in \Gamma(\mathcal{X} \setminus Z, \mathcal{O}_{\mathcal{X}}) \subset \mathcal{O}_{\mathcal{X},x}.$$

The image of f^{-1} under the local ring homomorphism $\mathcal{O}_{\mathcal{X},x} \to \mathcal{O}_v$ is a unit by assumption, hence f^{-1} is also a unit in $\mathcal{O}_{\mathcal{X},x}$. Consequently $f \in \mathcal{O}_{\mathcal{X},x}$.

□

We can now finish the proof of Proposition 1.28 i.

Proof of Proposition 1.28 i. Let $f \in \mathcal{O}_v^\times \subset F_X$ be such that the reduction $\overline{f} \in k(v)$ is transcendental over k. We also write $f\colon \mathcal{X} \dashrightarrow \mathbb{P}^1(\mathcal{O}_K)$ for the rational map induced by f, and let U be its domain of definition. Define $\Gamma_f \subset \mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}^1(\mathcal{O}_K)$ to be the graph of f. Recall that it is defined to be the closure of the graph $f|_U$ in $\mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}^1(\mathcal{O}_K) \supset U \times_{\mathcal{O}_K} \mathbb{P}^1(\mathcal{O}_K)$. This is almost a model of X, as it satisfies all the properties of a model except that it might not be normal.

$$\begin{tikzcd}
 & \mathcal{X}' \arrow[d] & \\
\mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}^1(\mathcal{O}_K) \arrow[r, phantom, "\supseteq"] & \Gamma_f \arrow[r] \arrow[d] & \mathbb{P}^1(\mathcal{O}_K) \\
 & \mathcal{X} \arrow[ur, dotted, "f"'] &
\end{tikzcd}$$

The first projection $\mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}^1(\mathcal{O}_K) \to \mathcal{X}$ induces a birational morphism $\Gamma_f \to \mathcal{X}$. The second projection induces a morphism $\Gamma_f \to \mathbb{P}^1(\mathcal{O}_K)$ which extends $f\colon \Gamma_f \dashrightarrow \mathbb{P}^1(\mathcal{O}_K)$. Let $\mathcal{X}' \to \Gamma_f$ denote the normalization of Γ_f. This gives a model of $\mathcal{X}'$ for which $f\colon \mathcal{X}' \to \mathbb{P}^1(\mathcal{O}_K)$ is a morphism. By Lemma 1.32 b, this implies that $v \in V(\mathcal{X}')$. □

1.2.2 Proof of the Second Part of Theorem 1.1

We fix a model $\mathcal{X}$ of X and a nonempty subset $V \subset V(\mathcal{X})$. Denote by $\mathcal{E}$ the set of irreducible (integral) components of $\mathcal{X}_s$ that induce the valuations in $V(\mathcal{X}) \setminus V$.

Definition 1.33. A model $\mathcal{X}'$ together with a modification $\varphi\colon \mathcal{X} \to \mathcal{X}'$ such that for every integral vertical curve E on $\mathcal{X}$, the set $\varphi(E)$ is a point if and only if $E \in \mathcal{E}$ is called a *contraction* of the $E \in \mathcal{E}$.

It can be shown that if a contraction exists, it is unique up to unique isomorphism. Our goal now is to show that a contraction of $\mathcal{E}$ exists. Once this is done, the proof is complete, since for the contraction $\varphi\colon \mathcal{X} \to \mathcal{X}'$, we have by construction $V(\mathcal{X}') = V$.

Lemma 1.34 ([**?**, Lemma 8.3.29, Lemma 8.3.3])**.** *Let $\mathcal{L}$ be an invertible sheaf on $\mathcal{X}$ generated by global sections $s_0, \dots, s_n$. Consider the morphism $f\colon \mathcal{X} \to \mathbb{P}^n(\mathcal{O}_K)$ associated to these sections. Let E be a vertical curve of $\mathcal{X}_s$ with the reduced induced closed subscheme structure. Then $f(E)$ is reduced to a point if and only if $\mathcal{L}|_E \cong \mathcal{O}_E$.*

Proposition 1.35 ([**?**, Proposition 8.3.30])**.** *The following are equivalent.*

(a) The contraction $\varphi\colon \mathcal{X} \to \mathcal{X}'$ of $\mathcal{E}$ exists.

(b) There exists a Cartier divisor D on $\mathcal{X}$ such that $\deg(D|_{\mathcal{X}_g}) > 0$, that $\mathcal{L} \cong \mathcal{O}_{\mathcal{X}}(D)$ is generated by its global sections, and that for any integral vertical curve E, we have $\mathcal{L}|_E \simeq \mathcal{O}_E$ if and only if $E \in \mathcal{E}$.

Proof. We only sketch (b) $\implies$ (a). For the other implication and details see [**?**, Proposition 8.3.30].

As $\mathcal{X}_g$ is an integral projective curve over a field, $\mathcal{L}|_{\mathcal{X}_g}$ is an ample divisor ([**?**, Proposition 7.5.5]). Replacing D by a positive multiple (which does not change the set E), we can suppose that $\mathcal{L}|_{\mathcal{X}_g}$ is very ample.

Let $f\colon \mathcal{X} \to \mathbb{P}^n(\mathcal{O}_K)$ be a morphism associated to sections $s_0, \dots, s_n \in \Gamma(\mathcal{X}, \mathcal{L})$ which generate $\mathcal{L}$. Let Z be the closed subset $f(\mathcal{X}) \subset \mathbb{P}^n(\mathcal{O}_K)$ endowed with the reduced (hence integral) scheme structure. Then f induces a dominant morphism $\mathcal{X} \to Z$ that we still denote by f. This is an isomorphism on the generic fibre because $\mathcal{L}|_{\mathcal{X}_g}$ is very ample. In particular, $f\colon X \to Z$ is birational. Denote by $\pi\colon \mathcal{X}' \to Z$ the normalization morphism. As $\mathcal{X}$ is normal, f uniquely factors into $\varphi\colon \mathcal{X} \to \mathcal{X}'$ followed by $\pi\colon \mathcal{X}' \to Z$.

$$\begin{tikzcd} & \mathcal{X}' \arrow[d, "\pi"] \\ \mathcal{X} \arrow[ur, "\varphi"] \arrow[r, "f"'] & Z \end{tikzcd}$$

Let E be an integral vertical curve on $\mathcal{X}$. Then we have

$$f(E) \text{ is a point } \iff \mathcal{L}|_E \simeq \mathcal{O}_E \iff E \in \mathcal{E}$$

where the first equivalence is a consequence of Lemma 1.34 and the second one is true by assumption. We claim that $\pi\colon \mathcal{X}' \to Z$ is a finite morphism. This then completes the proof because $f(E)$ is reduced to a point if and only if $\varphi(E)$ is a point. The morphism f is projective ([?, Corollary 3.3.32(e)]), so $\mathbb{A} \cong f_*\mathcal{O}_{\mathcal{X}}$ is a coherent sheaf on Z ([?, Corollary 5.3.5]). Furthermore, $\mathbb{A}$ is a sheaf of integrally closed algebras because $\mathcal{X}$ is normal. As a result, the normalization morphism $\pi\colon \mathcal{X}' \to Z$ coincides with the canonical morphism $\underline{\operatorname{Spec}}\,\mathbb{A} \to Z$ and is a finite morphism ([?, Exercise 5.1.17]). □

For the construction of a suitable Cartier divisor, we furthermore need the following technical lemma.

Lemma 1.36 ([?, Lemma 8.3.32]). *Let D be an effective horizontal Cartier divisor on $\mathcal{X}$, and $\mathcal{L} = \mathcal{O}_{\mathcal{X}}(D)$. Then there exists a $m_0 \geq 1$ such that $\mathcal{L}^{\otimes m}$ is generated by its global sections for every $m \geq m_0$.*

Proposition 1.35 and Lemma 1.36 show that for a contraction morphism to exist, it suffices that there exists an effective Cartier divisor that meets the closed fibres in a suitable way. To construct such a divisor, it is important that the ring $\mathcal{O}_K$ is complete, which implies it is Henselian.

Definition 1.37. Let A be a Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field $k := A/\mathfrak{m}$. We say that A is Henselian if one of the following equivalent conditions is fulfilled:

- Hensel's Lemma holds: For every monic $f \in A[T]$, and every simple root $a_0 \in k$ of $\overline{f} \in k[T]$, there exists a lift $a \in A$ (so $\overline{a} = a_0$) such that $f(a) = 0$.
- Every finite A-algebra is a direct sum of local A-algebras.

Lemma 1.38 ([?, 8.3.35]). *Let $x \in \mathcal{X}_s$ be a closed point on the special fiber, then the following assertions hold.*

(a) There exists an irreducible horizontal Weil divisor Δ containing x.

(b) There exists an effective horizontal Cartier divisor D such that $\mathcal{X}_s \cap D = \{x\}$.

Proof. (a) Let $\mathfrak{m}_x$ be the maximal ideal of $\mathcal{O}_{\mathcal{X},x}$ and denote by π an uniformizer of $\mathcal{O}_K$. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$ be the prime ideals of $\mathcal{O}_{\mathcal{X},x}$ that are minimal among those containing $\pi\mathcal{O}_{\mathcal{X},x}$. As $\dim_{\mathcal{O}_{\mathcal{X},x}} = 2$ ([?, Proposition 8.3.4]), Krull's principal ideal theorem implies that $\mathfrak{m}_x \neq \mathfrak{p}_i$. There therefore exists an

$$f \in \mathfrak{m}_x \setminus (\cup_i \mathfrak{p}_i).$$

Thus f is a regular element and $V(f)$ does not contain any irreducible component of $\operatorname{Spec} \mathcal{O}_{\mathcal{X}_s,x}$.

There exists an affine open subset $W \ni x$ and a regular element $g \in \mathcal{O}_{\mathcal{X}}(W)$ such that $g_x = f$, and that $V(g) \cap W_s = \{x\}$. Let Δ be the closure in $\mathcal{X}$ of an irreducible component of $V(g)$ passing through x. Then Δ is an irreducible horizontal divisor passing through x.

(b) Let $E := V(g)$ and let W be as in a. Then E can be seen as an effective Cartier divisor on W. Moreover, $E \to \operatorname{Spec} \mathcal{O}_K$ is quasi-projective and quasi-finite because $\dim_{\mathrm{E}_s} < \dim_{\mathcal{X}_s} = 1$. By a corollary of Zariski's Main theorem ([?, Corollary 4.4.8]), there exists an affine open neighborhood U of x such that $E \cap U$ is an open subscheme of a scheme Z that is finite over $\operatorname{Spec} \mathcal{O}_K$.

$$E \cap U \xhookrightarrow{\text{open emb.}} Z \xrightarrow{\text{finite}} \operatorname{Spec} \mathcal{O}_K$$

Let D be the connected component of Z containing x. Since $\mathcal{O}_K$ is a Henselian local ring, Z is a disjoint union of local schemes. Hence D is local, and consequently $D \subset E \cap U \subset \mathcal{X}$.

Finally, since D is closed in Z, it is finite over $\operatorname{Spec} \mathcal{O}_K$. A finite morphism $D \to \operatorname{Spec} \mathcal{O}_K$ between affine schemes is projective ([?, Exercise 3.3.22]), and therefore proper ([?, Theorem 3.3.30]). So the morphisms

$$D \to \mathcal{X} \to \operatorname{Spec} \mathcal{O}_K \quad \text{and} \quad \mathcal{X} \to \mathcal{O}_K$$

are proper, which implies that $D \to \mathcal{X}$ is also proper ([?, Proposition 3.3.16(e)]). Consequently, D is closed in $\mathcal{X}$ and therefore a Cartier divisor, as required.

□

Now we can finally prove Proposition 1.28 ii.

Proof of Proposition 1.28 (ii). Let $Z_1, \dots, Z_n$ be the irreducible components of $\mathcal{X}_s$ that induce the valuations in $V \subset V(\mathcal{X})$. By Lemma 1.38 b, there exists an effective horizontal Cartier divisor D_i such that $\mathcal{X}_s \cap D_i$ is a point of Z_i that belongs to none of the components of . Let $D = \sum_{i=1}^n D_i$. By virtue of Lemma 1.36, if necessary replacing D by a multiple, we can even suppose that $\mathcal{L} := \mathcal{O}_{\mathcal{X}}(D)$ is generated by its global sections.

The theorem then results from Proposition 1.35. Indeed, $\mathcal{L}|_E \simeq \mathcal{O}_E$ if $E \in$ since $E \cap D = \varnothing$; if $E \notin$, then E is some Z_i and $\deg \mathcal{L}|_E \geq \deg \mathcal{O}_{\mathcal{X}}(D_i)|_{Z_i} > 0$, and hence $\mathcal{L}|_E \not\simeq \mathcal{O}_E$. □

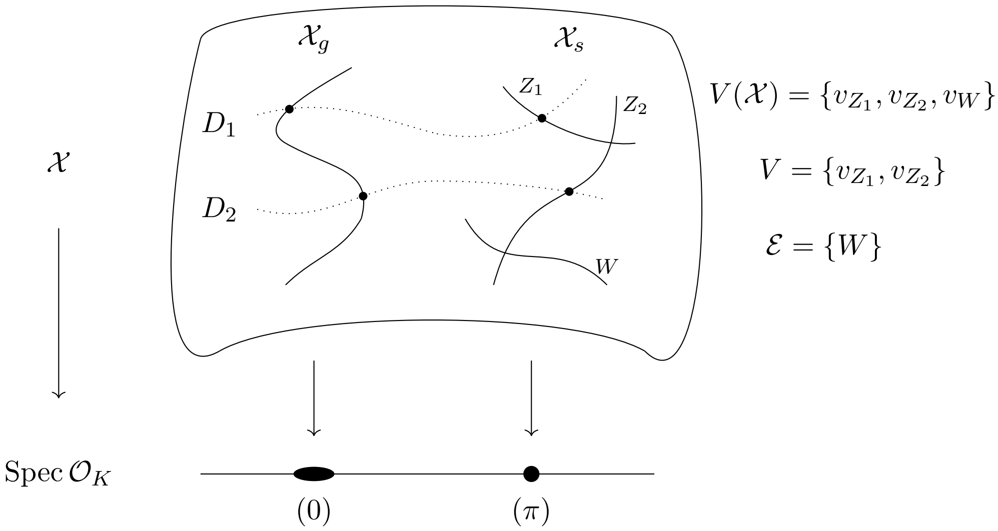

Figure 1.1: Illustration of the proof of Proposition 1.28 (ii)

Remark 1.39. The proof of Proposition 1.28 ii shows that actually all models $\mathcal{X}$ of X are projective. Indeed one just has to choose $V = V(\mathcal{X})$ in the proof to obtain an effective Cartier divisor D which meets all the irreducible components of the fiber $\mathcal{X}_s$. Since $\mathcal{X}_s$ is a projective curve ([**?**, Exercise 7.5.4]), $\mathcal{O}_{\mathcal{X}}(D)|_{\mathcal{X}_s}$ is ample ([**?**, Proposition 7.5.5]). It follows from [**?**, Corollary 5.3.24] that $\mathcal{O}_{\mathcal{X}}(D)$ is ample. So $\mathcal{X} \to \operatorname{Spec} \mathcal{O}_K$ is projective ([**?**, Proposition 5.3.6]).

1.3 Permament Models

1.3.1 Models with Reduced Special Fibre

As in the last two sections, we assume X to be a smooth, projective and absolutely irreducible curve over the local field K.

Definition 1.40. A model $\mathcal{X}$ of X is called *permanent*, if $\mathcal{X} \times_{\mathcal{O}_K} \mathcal{O}_L$ is normal for all finite field extensions $L|K$.

Remark 1.41. Since flatness and properness are preserved under base change, a model $\mathcal{X}$ of X is permanent if and only if $\mathcal{X} \times_{\mathcal{O}_K} \mathcal{O}_L$ is a model of $X_L = X \times_K L$.

Definition 1.42. A *pre-model* of X is a flat and proper $\mathcal{O}_K$-scheme $\mathcal{X}$ with $\mathcal{X} \times_{\mathcal{O}_K} K = X$. That is, $\mathcal{X}$ satisfies all the conditions of a model except of being normal.

Lemma 1.43 ([**?**, Lemma 1.3])**.** *Let $\mathcal{X}$ be a pre-model of X. Then $\mathcal{X}$ is normal (i.e. a model) if and only if the following two conditions hold.*

(i) *The special fibre $\mathcal{X}_s$ has no embedded points, i.e. no point $x \in \mathcal{X}_s$ satisfying $\mathfrak{m}_x = \mathrm{Ann}_R(r)$ for some $r \in R := \mathcal{O}_{\mathcal{X}_s,x}$ (see [?, Definition 7.1.6]).*

(ii) *The local ring $\mathcal{O}_{\mathcal{X}\eta}$ is a discrete valuation ring for every generic point $\eta \in \mathcal{X}_s$.*

Proof. According to Serre's criterion for normality [?, Theorem 8.2.23], we need to check the following properties:

(R_1) $\mathcal{X}$ is regular at x when $\operatorname{codim} x \leq 1$, and

(S_2) $\operatorname{depth} \mathcal{O}_{\mathcal{X},x} \geq 2$.

for all $x \in \mathcal{X}$.
The points on $\mathcal{X}$ with codimension 1 are precisely the points on the generic fibre, which is regular by assumption, and the generic point on the special fibre. Since any discrete valuation ring is a local Noetherian domain and vice versa, (R_1) is equivalent to (i). By [?, Proposition 8.2.11], condition (S_2) is equivalent to say that $\operatorname{depth} \mathcal{O}_{\mathcal{X}_s,x} \geq 1$ for all $x \in \mathcal{X}_s$. But this is always true for the generic point, and if x is a different point it is equivalent to say that x is not an embedded point. □

Example 1.44. 1. Consider a pre-model with special fibre

$$\mathcal{X}_s := \operatorname{Spec} k[x,y]/(x^2, xy), \tag{1.45}$$

which is topologically equivalent to $\mathbb{A}^1(k)$. The point corresponding to the maximal ideal (x, y) is an embedded point.

2. [?, Example 9.8.4].

Proposition 1.46 ([?, Proposition 2.3])**.** *If $\mathcal{X}_s$ is reduced, then $\mathcal{X}$ is permanent.*

Proof. Let $L|K$ be a finite extension and $l|k$ the extension of the corresponding residue fields. The special fibre of $\mathcal{X}' := \mathcal{X}_{\mathcal{O}_L} = \mathcal{X} \times_{\mathcal{O}_K} \mathcal{O}_L$ is $\mathcal{X}'_s := \mathcal{X}_s \times_k l$, which is reduced since k is perfect by assumption. This means $\mathcal{X}'_s$ satisfies condition (ii) from Lemma 1.43.
For condition (i) let $\eta \in \mathcal{X}'_s$ be any generic point. The chalk $\mathcal{O}_{\mathcal{X}',\eta}$ then is a Noetherian local domain of dimension 1 and it holds

$$\mathcal{O}_{\mathcal{X}'_s,\eta} = \mathcal{O}_{\mathcal{X}',\eta}/\pi_L \mathcal{O}_{\mathcal{X}',\eta}, \tag{1.47}$$

where $\pi_L \in \mathcal{O}_L$ is a uniformising parameter. Since $\mathcal{X}'_s$ is reduced, $\pi_L \mathcal{O}_{\mathcal{X}',\eta}$ is maximal ideal which means that condition (i) from Lemma 1.43 is satisfied as well. Thus $\mathcal{X}'$ is normal and hence $\mathcal{X}$ is permanent. □

Proposition 1.48 ([**?**, Corollary 2.2 (ii)]). *Let $\mathcal{X}$ be a model of X, then the following statements are equivalent.*

(i) *$\mathcal{X}_s$ is reduced.*

(ii) *$\mathcal{O}_{\mathcal{X}_s,\eta}$ is a field for every generic point $\eta \in \mathcal{X}_s$.*

(iii) *Every $v_Z \in V(\mathcal{X})$ is weakly ramified.*

Proof. Since $\mathcal{X}$ is normal, there are no embedded points in $\mathcal{X}_s$ by Lemma 1.43. Since $\mathcal{X}_s$ has dimension one, it is reduced if and only if $\mathcal{X}_s$ is reduced in codimension zero which in turn is the case if $\mathcal{O}_{\mathcal{X}_s,\eta}$ is a field for every generic point $\eta \in \mathcal{X}$.
For the second equivalence recall that according to Proposition 1.18, for any generic point $\eta \in \mathcal{X}_s$ and $Z = \overline{\{\eta\}}$ it holds

$$[\Gamma_{v_Z} : \Gamma_K] = \operatorname{length} \mathcal{O}_{\mathcal{X}_s,\eta}. \tag{1.49}$$

This implies that $\mathcal{O}_{\mathcal{X}_s,\eta}$ is a field if and only v_Z is weakly ramified for all $v_Z \in V(\mathcal{X})$. □

Example 1.50 (Curve with reduced special fibre). Let $K = \mathbb{Q}_3$ and consider the curve

$$X : y^3 = 3x^2 + 1. \tag{1.51}$$

The equation is already defined over $\mathcal{O}_K = \mathbb{Z}_3$, i.e. we can define a model with the same equation. The reduction $\mathcal{X}_s/\mathbb{F}_3$ is given by $0 = \bar{y}^3 - 1 = (\bar{y}-1)^3$ which means $\mathcal{X}_s$ is not reduced.
The irreducible component is given by $Z : y - 1$; denote the corresponding valuation by v. For $z := y - 1$ we have

$$\begin{aligned} 3x^2 &= y^3 - 1 \\ &= (z+1)^3 - 1 \\ &= z^3 + 3z^2 + 3z, \end{aligned}$$

where the right hand side is an Eisenstein polynomial. This results $v(z) = \frac{1}{3} v_G(3x^2) = \frac{1}{3} \notin \mathbb{Z} = \Gamma_K$. Let us extend the ground field to $L = \mathbb{Q}_3[\theta]$ where $\theta^3 - 3 = 0$ and apply the coordinate transformation $(x, y) \mapsto (x_1, 1 + \theta y_1)$. We then obtain a birational curve

$$X_1 : y_1^3 + \theta^2 y_1^2 + \theta y_1 = x_1^2. \tag{1.52}$$

Since $y_1^3 \equiv x_1^2 \pmod{\theta}$, $\mathcal{X}_s$ is reduced. It holds $v_G(\theta y_1) = v_G(y - 1) = \frac{1}{3} = v_L(\theta) \in \Gamma_L$, especially $v_G(y_1) = 0$.

1.3.2 Models under Base Field Extensions

Let $L|K$ be a finite field extension and let $\mathcal{X}$ be a model of X. Denote by $\mathcal{X}' := (X \times_{\mathcal{O}_K} \mathcal{O}_L)^{\sim}$ the normalised base change of $\mathcal{X}$ to $\mathcal{O}_L$. Let $\eta \in \mathcal{X}_s$ be a generic point and let $\eta' \in \mathcal{X}'_s$ be a generic point lying over η. Since $\mathcal{X}$ (resp. $\mathcal{X}'$) is normal we have that $R := \mathcal{O}_{\mathcal{X},\eta}$ (resp. $:= \mathcal{O}_{\mathcal{X}',\eta'}$) is a discrete valuation ring dominating $\mathcal{O}_K$ (resp. $\mathcal{O}_L$) by Lemma 1.43 (i), that is

- $R \supset \mathcal{O}_K$ (resp. $S \supset \mathcal{O}_L$), and
- $\mathfrak{m}_R \cap \mathcal{O}_K = \pi_K \mathcal{O}_K$ (resp. $\mathfrak{m}_S \cap \mathcal{O}_L = \pi_L \mathcal{O}_L$).

We arrive at the following diagram of fields.

$$\begin{tikzcd}
 & (F_{X_L}, \{w_{i,1}, \dots, w_{i,j_i}\}) \arrow[dd, dash] \\
(F_X, \{v_i\}) \arrow[ur, dash] \arrow[dd, dash] & \\
 & (L(x), v_{G,L}) \arrow[dd, dash] \\
(K(x), v_{G,K}) \arrow[ur, dash] \arrow[dd, dash] & \\
 & (L, v_L) \\
(K, v_K) \arrow[ur, dash] &
\end{tikzcd}$$

Here the v_i are the valuations on F_X corresponding to the generic points of the components of $\mathcal{X}_s$. The transcendental element of the field extension $K(x)|K$ can be chosen such that the Gauß valuation $v_{G,K}$ on $K(x)$ is compatible with the extension of valuations.
$w_{i,j}$ represent the valuations corresponding to the irreducible component of $\mathcal{X}'_s$ whose generic point lies over the generic point of $Z_i \subset \mathcal{X}_s$ which corresponds to v_i.

Theorem 1.2 ([**?**, Proposition 2.3])**.** *Let $\mathcal{X}$ be a model of X. Then there is a finite extension $L|K$ such that $\mathcal{X}' = (\mathcal{X} \times_{\mathcal{O}_K} \mathcal{O}_L)^{\sim}$ has a reduced special fibre.*

Proof. By [**?**, Theorem 1.0] there exists a field extension $L|K$ such that $w_{i,j}$ is weakly unramified over v_L. According to Proposition 1.48, the special fibre of $\mathcal{X}'$ is reduced. □

Lemma 1.53 (Abhyanka)**.** *Let $L|K$ be a finite extension. Let $\{v_1, \dots, v_n\} = V(X)$ denote the extended valuation on the function field $F_X|K$. We have* *that $F_{X_L} = F_X L$ and with the extended valuation denoted as $w_{1,1}, \dots, w_{n,j_n}$ respectively. Assume that $v_i | v_{G,K}$ are tamely ramified for all i. If $e_i \coloneqq e(v_i | v_{G,K})$ divides $e(v_{G,L}, v_{G,K}) = e(v_L | v_K)$ for all i, then $w_{i,j} \in V(X_L)$ is weakly unramified over $v_{G,L}$ for all i, j.*

Proof. We will use the equivalence of Proposition 1.48 without further mention, that is the equivalence of the special fibre being reduced and the valuations being mildly ramified. Without loss of generality we may assume the following assertions:

- $n = 1$ and $j_1 = 1$ since the argument can be constructed inductively,
- $e \coloneqq e_1 = e(v_1 | v_{G,K}) = e(v_{G,L} | v_{G,K})$ since we assume the residue field to be perfect, and
- K contains the e-th root of unity, i.e. $\pi_K = \pi_K^e$, since we can always replace K by a suitable Kummer extension without violating the other assumptions and obtain a compatible diagram regarding the valuations.

Since $F_{X_L} = F_X L$ we have that $[\Gamma_{w_1} : \Gamma_{v_1}] = 1$. By multiplicativity of the ramification indices we conclude $[\Gamma_{w_1} : \Gamma_{v_L}] = 1$. □

Example 1.54 (Reduction after wildly ramified field extension)**.** Consider the curve

$$X : y^2 = 8x^2 - 2 \tag{1.55}$$

defined over the field $K \coloneqq \mathbb{Q}_2$. This example illustrates that the right choice of the field extension is crucial for having a reduced special fibre, since a wild ramified field extension is not unique.

(i) First let us consider the extension $L_1 = K(\theta_1)$ where $\theta_1^2 - 2 = 0$ and the coordinate transformation $(x, y) \mapsto (x_1, 2y_1 - \theta_1 - 2)$. We obtain a birational curve

$$X_1 : y_1^2 - (\theta_1 + 4)y_1 = 2x_1^3 - (2 + \theta_1) \tag{1.56}$$

with $v_{G,L_1}(y_1) = \frac{1}{4}$. We have $[\Gamma_{F_{X_1}} : \Gamma_{L_1}] = 2 \neq 1$.

(ii) Let us consider the field extension $L_2 = K(\theta_2)$ where $\theta_2^2 + 2 = 0$ and the coordinate transformation $(x, y) \mapsto (x_2, 2\theta_2 y_2 + \theta_2)$. We obtain a birational curve

$$X_2 : y_2^2 + y_2 = -x_2^3 \tag{1.57}$$

which is given by an equation of AS-form, i.e. its special fibre is a reduced curve. Thus $[\Gamma_{F_{X_2}} : \Gamma_{L_2}] = 1$.